

CONTEXT-CONDITIONAL ADAPTIVE DISTILLATION: EQUALIZING TEACHER MARGIN DISTRIBUTIONS FOR ROBUST AND STABLE KNOWLEDGE TRANSFER

Anonymous authors

Paper under double-blind review

ABSTRACT

We study a pervasive but under-addressed failure mode in knowledge distillation (KD): the teacher’s decision margin varies systematically with input context, defined by augmentation states (brightness, contrast, scale, translation, blur) and latent style, which induces context-conditional distribution mismatch. Students inherit this skew, degrading cross-architecture transfer, performance on long-tail and rare-context cells, calibration, and worst-group robustness. Student-driven reweighting can be unstable early [niu₂₀₂₂, *especting*, zhou₂₀₂₁, *ethinking*], global temperature schedules ignore contextual sharpness Zheng & Yang (2024). We introduce Context-Conditional Adaptive Distillation (CCAD), a teacher statistics-only, plug-in mechanism that discovers contexts without labels, builds per-context empirical CDFs (ECDFs) of a boundary distance proxy, and uses quantiles to drive three coupled controls: rank, stable intra-context weights, inter-context distributionally robust (DRO) group weights, and a context adaptive teacher temperature. An optional per context affine logit transform aligns margin distributions to a global reference. CCAD integrates into standard stages of training for early stability and zero test time overhead. We provide complete PyTorch utilities and a rigorous evaluation plan on CIFAR-100 and ImageNet-100 with robustness suites, ablations, and fairness safeguards. Our shared execution artifacts, register hypotheses and analysis protocols to enable conclusive verification.

1 INTRODUCTION

Knowledge distillation (KD) transfers a pretrained teacher’s predictive behavior to a student by matching soft targets, often improving generalization and efficiency. Yet KD frequently overlooks that the teacher’s confidence varies with input context, including explicit augmentation states (brightness, contrast, saturation, scale, translation, blur) and latent style captured by self-supervised features. In practice, the teacher’s decision margin distribution shifts across such contexts. When the student learns from these context-skewed margins, it inherits the skew: performance deteriorates on rare context $\times$ class cells, calibration degrades, and worst-group accuracy declines under strong augmentation and out-of-distribution (OOD) shifts [ojha₂₀₂₂, *what*, cho₂₀₁₉, *the*].

This problem is hard for four reasons. First, imbalance is context-conditional: the same absolute decision difficulty can be over- or under-represented depending on a context’s margin distribution. Second, early training is fragile; student-driven reweighting can be unstable when the student’s predictions are still nascent [niu₂₀₂₂, *especting*, zhou₂₀₂₁, *ethinking*]. Third, temperature scaling—central to KD—has been largely applied as a global varying schedule that ignores contextual sharpness Zheng & Yang (2024). Fourth, although ECDF-based methods can correct shifts without density ratio estimation Mazaheri et al. (2020), they have not been used to equalize dependent margin distributions within KD. Parallel lines of work emphasize invariance and worst-group robustness, yet KD rarely optimizes worst-group risk explicitly Arjovsky et al. (2019).

We argue that robust KD should concurrently: (1) equalize the teacher’s boundary-distance distributions across discovered contexts, (2) adapt teacher sharpness to context difficulty, and (3) explicitly protect worst-group accuracy. We introduce Context-Conditional Adaptive Distillation (CCAD), a teacher-only, rank-based mechanism that operationalizes these principles with minimal overhead and zero test-time cost. CCAD discovers contexts without labels by combining augmentation logs with unsupervised style clustering; maintains per-context ECDFs of a boundary-distance proxy (top-1 minus top-2 logit margin by default); and uses quantiles to drive three coupled controls: rank-stable intra-context weights, inter-context distributionally robust (DRO) group weights using exponentiated-gradient updates on per-context moving-average KD loss, and a context-adaptive teacher temperature. An optional per-context affine transform aligns per-context margin ECDFs to a global reference. A two-stage schedule collects ECDFs during warm-up before ramping controls, yielding early stability. CCAD is orthogonal to feature-level and information-theoretic transfer and can be layered with them [chung2020_{feature},ahn2019_{variational}].

We evaluate CCAD on CIFAR-100 and ImageNet-100 with cross-architecture pairs (e.g., ViT to ResNet/MobileNet), robustness on ImageNet-C and Stylized-ImageNet, worst-group accuracy over discovered contexts, tail-quantile accuracy, calibration under shift, and early-stability diagnostics. Baselines include vanilla KD, TTM/WTTM Zheng & Yang (2024), inverse-propensity and weighted soft-label methods [niu2022_{respecting},zhou2021_{ethinking}], and a GroupDRO – only variant. We include ablations for each CCAD component and sensitivity to key hyperparameters, enforcing fairness tests. Our execution artifact validates send-to-end integration and measurement but contains nonnumerical results; we register analysis plans.

Contributions:

- **Context-conditional mismatch.** We formalize context-conditional distribution mismatch in KD and propose a teacher-statistics-only solution based on per-context ECDFs to equalize boundary-distance distributions.
- **Rank-based control triad.** We introduce a triad of rank-based controls: (A) intra-context weights that are rank-invariant, (B) inter-context DRO weights that optimize worst-group performance, and (C) a context-adaptive teacher temperature that prevents transfer of context skew; plus an optional quantile-equalized teacher.
- **Stable two-stage design.** We design a two-stage schedule and safety mechanisms ensuring early stability, rare-context smoothing, and zero test-time overhead, with minimal training overhead.
- **Evaluation plan and utilities.** We provide a comprehensive evaluation plan with baselines, ablations, robustness, calibration, and stability metrics, along with open-source PyTorch utilities to reproduce results.

Beyond classification, CCAD naturally extends to detection and segmentation by treating regions as contexts, and to multimodal KD by leveraging environmental embeddings. It complements broader efforts in dataset distillation and anytime adaptation [author_{year}_{large},author_{year}_{anyda}], causal interventions for distillation aut (b), multi – level logit transfer aut (e), and auxiliary – variable formulations aut (c).

2 RELATED WORK

Soft-label distillation and temperature scaling. KD aligns student predictions to softened teacher outputs; the bias-variance trade-off induced by soft labels motivates sample-adaptive weighting Zhou et al. (2021). Transformed Teacher Matching (TTM) and Weighted TTM (WTTM) reinterpret temperature scaling as a power transform with an inherent Rényi-entropy regularizer, improving generalization while keeping the student temperature fixed Zheng & Yang (2024). However, they use global or time-varying temperatures and cannot adapt to context-specific sharpness. CCAD complements these approaches by setting teacher temperature as a function of per-context quantiles while keeping the student temperature fixed, consistent with TTM’s philosophy.

Reweightings under transfer gaps. Inverse Probability Weighting Distillation (IPWD) addresses non-IID transfer by estimating a propensity score and inversely weighting samples Niu et al. (2022). Weighted soft labels adopt sample-wise weighting to balance bias-variance Zhou et al. (2021). Such

strategies depend on student-side signals and can destabilize early optimization. CCAD is teacher-only and rank-based, avoiding dependence on nascent student predictions and explicitly balancing contexts. Related work in unbiased learning-to-rank jointly learns propensity and ranking models Ai et al. (2018), echoing our integration of group weighting into KD; in contrast, we operate over discovered contexts and teacher–student divergences.

Distribution shift and ECDF-based correction. ECDF-based methods correct covariate shift without explicit density ratio estimation and tend to be robust to hyperparameters Mazaheri et al. (2020). We adapt this idea to KD by using per-context ECDFs of teacher margins to equalize internal distributions across contexts rather than reweighting between datasets. This within-task, per-context usage removes the need for target density estimation and enables stable rank-based controls tailored to KD.

Worst-group robustness and invariance. Invariant Risk Minimization (IRM) seeks representations for which the optimal classifier is stable across environments Arjovsky et al. (2019). Our goal is complementary: ensure distributional balance and protect worst-group accuracy across discovered contexts via inter-context DRO updates driven by teacher–student divergences. This sidesteps strong invariance assumptions and does not require labeled environments.

Feature-level and information-theoretic transfer. Online adversarial distillation aligns feature-map distributions Chung et al. (2020), and variational information distillation maximizes mutual information between teacher and student Ahn et al. (2019). CCAD operates at the logit level, adding negligible overhead, and is orthogonal to these methods, enabling straightforward composition.

Domain adaptation and black-box predictors. KD3A performs decentralized multi-source adaptation via distillation with consensus mechanisms Feng et al. (2020), while BETA mitigates confirmation bias in black-box adaptation through divide-to-adapt and mutual teaching Yang et al. (2022). These works target cross-domain adaptation, whereas CCAD addresses single-teacher KD with context-conditional imbalance within the training distribution. Comprehensive KD with causal interventions aut (b), multi-level logit distillation aut (e), dataset distillation under domain shift aut (d), anytime adaptation aut (a), auxiliary-variable KD aut (c), and analyses of teacher–student capacity mismatch Cho & Hariharan (2019) offer complementary design dimensions and motivation.

Summary of contrasts. Compared with IPWD/WSL [niu2022_r,especting,zhou2021_r,ethinking], CCAD avoids student–dependence and adds explicit worst–group control. Compared with TTM/WTM Zheng & Yang (2024), CCAD adapts to based shift correction Mazaheri et al. (2020), CCAD equalizes teacher margin distributions internally to KD rather than the theoretic KD [chung2020_{feature},ahn2019_v,ariational], CCAD is lightweight and complementary.

3 BACKGROUND

Problem setting. We consider supervised K -way classification with a fixed teacher f_t and a student f_s trained via KD. Each input x has an unlabeled context t that captures augmentation states (brightness, contrast, saturation, crop scale, translation, blur) and latent style discovered via unsupervised clustering of CLIP/DINO features. The teacher produces logits $y_t(x) \in \mathbb{R}^K$. We define a boundary-distance proxy $m(x, t)$, by default the logit margin given by the difference between the highest and second-highest logits. Empirically, the distribution of m depends on t , creating context-conditional sharpness skew that propagates to the student.

Notation and ECDFs. For each context t , we maintain an empirical cumulative distribution function F_t over m , and a global reference F_{all} aggregated across all contexts. Given x in context t , we compute quantiles $q_t = F_t(m(x, t))$ and $q_{\text{ref}} = F_{\text{all}}(m(x, t))$. These rank statistics are stable under monotone transformations and avoid assumptions on absolute margin scales. CCAD uses q_t and q_{ref} to drive three controls: (i) an intra-context weight $w_{\text{ctx}}(x, t)$ that redistributes gradient mass within t , (ii) an inter-context group weight α_t that emphasizes high-loss contexts, and (iii) a context-adaptive teacher temperature $T_t(x)$ that modulates sharpness by difficulty. Optionally, we apply a per-context affine transform $s_t y_t(x) + b_t$ to align each context’s margin distribution to the global reference.

Objective. Let p_t^T denote the teacher’s softmax at temperature T and p_s the student’s softmax. CCAD augments standard KD with rank-based weights and group DRO, yielding the loss

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \alpha \sum_t \alpha_t \mathbb{E}_{x \in t} \left[w_{\text{ctx}}(x, t) \text{KL}(p_t^{T_t(x)} \| p_s(x)) \right].$$

During warm-up, $w_{\text{ctx}} = 1$, α_t is uniform, and $T_t(x) = T_{\text{base}}$ while ECDFs are collected. Thereafter, the intra-context exponent β , the DRO learning rate for α_t , and the temperature range are ramped with a cosine schedule for stability.

Assumptions and scope. CCAD requires no labels to discover contexts and uses only teacher statistics during warm-up. It assumes logit margins are informative about confidence, a standard proxy in KD and calibration. Rare contexts are smoothed via minimum-count thresholds for ECDF queries, hierarchical backoff to nearest style clusters by feature similarity, or fall back to the global pool. CCAD is compatible with transformed-teacher KD Zheng & Yang (2024), with reweighting methods [niu2022, respecting, zhou2021, rethinking], and with feature – level or information – theoretic KD [chung2020, feature, ahn2019, variational]. Foundational ideas in causal invariance Arjovsky et al. (2019), based, teacher – only design.

4 METHOD

4.1 OVERVIEW AND COMPONENTS

CCAD is a teacher-only, rank-based framework that equalizes teacher margin distributions across contexts, protects worst-group accuracy via group DRO, and adapts teacher sharpness to context difficulty. It consists of: (1) unsupervised context discovery, (2) per-context ECDF construction, (3) three quantile-driven controls, (4) an optional logit-side quantile equalization, and (5) a two-stage schedule with safety mechanisms.

4.2 CONTEXT DISCOVERY WITHOUT LABELS

Known contexts are derived from augmentation logs by discretizing brightness, contrast, saturation, crop scale, translation, and blur into fixed bins and concatenating bin identifiers. Latent style contexts are discovered by extracting CLIP or DINO features, applying PCA, and clustering with mini-batch k-means; cluster IDs serve as style contexts. A hybrid identifier concatenates augmentation bins with the style cluster. For sparse cells we use hierarchical backoff: merge to the nearest style cluster by cosine similarity or fall back to the global pool to guarantee sufficient support.

4.3 ECDFs AND RANK-STABLE INTRA-CONTEXT WEIGHTING

For each context t , maintain an online quantile sketch of $m(x, t)$ to form F_t ; maintain F_{all} globally. For a sample x in context t , compute $q_t = F_t(m(x, t))$ and $q_{\text{ref}} = F_{\text{all}}(m(x, t))$. Define the intra-context weight

$$w_{\text{ctx}}(x, t) = \text{clip}_{[c_{\min}, c_{\max}]} \left(\left(\frac{q_{\text{ref}} + \delta}{q_t + \delta} \right)^\beta \right).$$

Intuition: for the same absolute margin m , samples that lie in the tail of a biased context (lower q_t relative to q_{ref}) receive larger weights, equalizing gradient mass across contexts. Because it uses ranks, w_{ctx} is invariant to monotone rescaling and depends only on teacher signals.

4.4 INTER-CONTEXT DISTRIBUTIONALLY ROBUST WEIGHTS

Maintain a moving average $\hat{\mathcal{L}}_t$ of per-context KD losses. Update group weights by an exponentiated-gradient rule

$$\alpha_t \propto \exp(\lambda(\hat{\mathcal{L}}_t - \bar{\mathcal{L}})), \quad \text{then normalize and clip to } [a_{\min}, a_{\max}],$$

where $\bar{\mathcal{L}}$ is the mean over contexts. This DRO-style reweighting increases emphasis on contexts with larger recent loss, explicitly optimizing worst-group performance while remaining teacher-driven and early-stable.

4.5 CONTEXT-ADAPTIVE TEACHER TEMPERATURE

Set teacher temperature per sample by context quantile

$$T_t(x) = T_{\min} + (T_{\max} - T_{\min})(1 - q_t)^\gamma.$$

Low-quantile (hard-tail) samples receive higher temperatures, attenuating overconfidence and preventing the transfer of context-skewed sharpness. The student temperature remains fixed, preserving compatibility with transformed-teacher matching Zheng & Yang (2024).

4.6 OPTIONAL QUANTILE-EQUALIZED TEACHER

Learn per-context affine transforms $y \mapsto s_t y + b_t$ to align each context’s margin ECDF to the global reference, using a lightweight Wasserstein-1 proxy computed on a small buffer, with strong regularization and exponential moving averages. Well-matched contexts keep $s_t \approx 1$ and $b_t \approx 0$. This neutralizes teacher-side context bias at the source without changing the KD loss form.

4.7 LOSS AND TRAINING SCHEDULE

The per-batch objective is

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \alpha \sum_t \alpha_t \mathbb{E}_{x \in t} \left[w_{\text{ctx}}(x, t) \text{KL}(p_t^{T_t(x)} \| p_s(x)) \right].$$

Two-stage stability: during warm-up, collect ECDFs with $w_{\text{ctx}} = 1$, α_t uniform, and $T_t(x) = T_{\text{base}}$. Then ramp β , λ , and $(T_{\text{max}} - T_{\text{min}})$ with cosine schedules; enable the optional affine quantile-equalization after ECDFs stabilize. In late fine-tuning, optional small factors based on student-driven signals (IPWD to the power η and weighted soft labels to the power $1 - \eta$ with η around 0.2) can be applied cautiously for incremental gains while preserving early stability [niu2022,especting, zhou2021,ethinking].

4.8 RELIABILITY AND IMPLEMENTATION

Rare-context smoothing uses minimum-count thresholds for ECDF queries, hierarchical blending across nearest contexts in feature space, and backoff to F_{all} when necessary. Safety measures include clipping w_{ctx} and α_t (e.g., to $[0.25, 4]$), gradient-norm clipping, and constraining T within $[2, 8]$. Overhead is modest: one quantile sketch per context with logarithmic query time and a few tensor operations per batch. A ContextManager maps samples to context IDs; an ECDF store returns q_t and q_{ref} ; group weights α_t are updated from per-context losses; $T_t(x)$ and w_{ctx} are computed per sample; the optional affine transform applies per-context to teacher logits. There is no test-time overhead.

4.9 RELATIONSHIP TO PRIOR DESIGNS

Compared with IPWD/WSL, CCAD is teacher-only and rank-based, which enables early stability and explicit context balancing [niu2022,especting, zhou2021,ethinking]. Compared with TTM/WTM, CCAD makes temperature context-conditional Zheng & Yang (2024). Compared with ECDF-based shift correction Mazaheriet al. (2020), CCAD equalizes level and information – theoretic approaches [chung2020,feature, ahn2019,variational].

5 EXPERIMENTAL SETUP

We design three complementary evaluations to test cross-architecture transfer, worst-group robustness under discovered contexts, calibration, OOD robustness, and early stability. Implementations use PyTorch with mixed precision and gradient clipping. We keep data pipelines, schedules, and optimizer settings identical across methods unless required by a baseline, and we ensure zero test-time overhead for CCAD.

5.1 SHARED COMPONENTS AND SAFEGUARDS

Context discovery combines augmentation logs with style clustering (CLIP or DINO features, PCA, MiniBatchKMeans). ECDFs use per-context and global quantile sketches with minimum-count thresholds; rare contexts back off to the global ECDF. Intra-context weights use δ about 10^{-3} and β ramped by cosine; inter-context DRO uses exponentiated-gradient updates on per-context EMA losses, clipped to $[0.25, 4]$; $T_t(x)$ uses T_{min} between 2 and 2.5, T_{max} between 7 and 8, and γ between 1.0

Algorithm 1 CCAD training with two-stage schedule

```

1: Initialize student parameters, context manager, ECDF store (per-context and global), group
   weights  $\alpha_t \leftarrow 1/|\mathcal{T}|$ 
2: Set warm-up epochs  $E_w$ , base temperature  $T_{\text{base}}$ ; set schedules for  $\beta, \lambda, T_{\text{min}}, T_{\text{max}}$ 
3: for epoch = 1 to  $E$  do
4:   for minibatch  $\{x_i, y_i\}_{i=1}^B$  do
5:     Assign context  $t_i \leftarrow \text{ContextManager}(x_i)$ 
6:     Compute teacher logits  $\ell_i$  and student logits  $s_i$ 
7:     Compute margin  $m_i$  from  $\ell_i$  (top-1 minus top-2 logit)
8:     Update ECDF sketches with  $m_i$  per context  $t_i$  and globally
9:     if epoch  $\leq E_w$  then
10:      Set  $q_{t_i}, q_{\text{ref},i}$  via ECDFs; set  $w_i \leftarrow 1, T_i \leftarrow T_{\text{base}}$ 
11:     else
12:      Query quantiles  $q_{t_i} \leftarrow F_{t_i}(m_i), q_{\text{ref},i} \leftarrow F_{\text{all}}(m_i)$ 
13:      Compute intra-context weight  $w_i \leftarrow \text{clip}\left(\left(\frac{q_{\text{ref},i} + \delta}{q_{t_i} + \delta}\right)^\beta, [c_{\text{min}}, c_{\text{max}}]\right)$ 
14:      Compute temperature  $T_i \leftarrow T_{\text{min}} + (T_{\text{max}} - T_{\text{min}})(1 - q_{t_i})^\gamma$ 
15:      Optionally apply affine transform to  $\ell_i$ :  $\ell_i \leftarrow s_{t_i} \ell_i + b_{t_i}$ 
16:     end if
17:     Compute teacher distribution  $p_i \leftarrow \text{softmax}(\ell_i/T_i)$ , student distribution  $q_i \leftarrow$ 
      softmax( $s_i$ )
18:     Accumulate per-context KD losses  $\mathcal{L}_{\text{KD}}^{(t)} \leftarrow \sum_{i:t_i=t} w_i \text{KL}(p_i \| q_i)$ 
19:     Compute batch loss  $\mathcal{L} \leftarrow \mathcal{L}_{\text{ce}} + \alpha \sum_t \alpha_t \mathcal{L}_{\text{KD}}^{(t)} / N_t$ 
20:     Update student parameters by SGD/AdamW on  $\nabla \mathcal{L}$ ; apply gradient clipping
21:   end for
22:   if epoch  $> E_w$  then
23:     Update  $\hat{\mathcal{L}}_t$  as EMA of per-context KD losses
24:     Update group weights:  $\alpha_t \propto \exp(\lambda(\hat{\mathcal{L}}_t - \bar{\mathcal{L}}))$ ; normalize and clip
25:     Advance cosine schedules for  $\beta, \lambda, T_{\text{min}}, T_{\text{max}}$ 
26:   end if
27: end for

```

and 1.2. The optional quantile-equalized teacher employs strong regularization with exponential moving averages. Safety includes clipping all per-sample weights and temperatures, gradient-norm clipping, capping tracked contexts (for example, 400–1024 most frequent), and seed control. We run multiple seeds and report mean $\pm$ standard deviation with paired t-tests for worst-group accuracy and ECE.

5.2 EXPERIMENT 1: CIFAR-100 CROSS-ARCHITECTURE KD WITH DISCOVERED CONTEXTS

Dataset: CIFAR-100 with strong training augmentation and standard test. Teacher: WideResNet-70-16 or ViT-S/16 with top-1 above 80 percent. Student: ResNet-18 or MobileNetV2. Contexts: augmentation bins over brightness, contrast, and crop scale (three bins each) and style clusters via CLIP features with PCA dimension 64 and $k = 8$, producing up to 216 hybrid cells with hierarchical backoff for sparsity. Boundary-distance proxy: teacher margin (top-1 minus top-2 logit gap). Training: 200 epochs, batch size 512, SGD with momentum 0.9, weight decay 5×10^{-4} , cosine learning rate schedule. Warm-up for 10 epochs (collect ECDFs, $T = 4$, $w_{\text{ctx}} = 1$, α_t uniform). Then ramp β from 0 to 1.5, the DRO update strength from 0 to about 0.2, and T_{max} from 4 to 7; $T_{\text{min}} = 2$, $\gamma = 1.2$. KD weight $\alpha = 1.0$ with cross-entropy enabled. Metrics: top-1, negative log-likelihood (NLL), ECE with 15 bins, worst-group accuracy defined as the minimum accuracy among hybrid contexts with at least 100 test samples, tail-quantile accuracy at $q_t \leq 0.1$ (quantiles computed using frozen ECDFs), early stability via KD-loss variance and gradient-norm coefficient of variation (epochs 1–20), and per-epoch time overhead. Baselines: vanilla KD (fixed $T = 4$), TTM/WTM Zheng & Yang (2024), IPWD and weighted soft labels [niu2022, *respecting*, zhou2021, *ethinking*], and CCAD ablations (w_{ctx} only, α_t only, T_t only, with and without the optional affine equalization).

5.3 EXPERIMENT 2: IMAGENET-100 ROBUSTNESS AND OOD

Dataset: ImageNet-100 subset. Teacher: ViT-B/16 pretrained on ImageNet-1K. Students: ResNet-50 and MobileNetV3-Large. Contexts: augmentation bins over brightness, contrast, saturation, and crop scale (three bins each) and style clusters via CLIP features with PCA dimension 128 and $k = 16$. Cap the top 400 contexts by frequency; rare contexts fall back to the global ECDF. Training: 120 epochs, warm-up 5 epochs, batch size 1024 with gradient accumulation if needed, AdamW with learning rate 3×10^{-4} cosine schedule and weight decay 10^{-4} , label smoothing 0.1. Hyperparameters: β up to 1.0, DRO scaling up to 1.5, $T_{\text{min}} = 2$, $T_{\text{max}} = 8$, $\gamma = 1.2$. Enable the optional affine equalization after epoch 10. Baselines: vanilla KD, TTM/WTM, IPWD, weighted soft labels, and a GroupDRO-only variant that uses α_t over discovered contexts but no ECDF weights or adaptive temperature. Metrics: clean top-1/top-5, ECE, NLL; robustness on ImageNet-C with mean corruption error across types and severities (mCE), corruption ECE (ECE-C), mean flip rate across severities (mFR), AUROC for confidence versus error on corruptions; worst-group accuracy over hybrid contexts on clean validation and over frequent corruption types at severity 5; calibration shift as the difference between clean ECE and average ECE at severity 5; and cross-architecture deltas (ViT to ResNet-50 versus ViT to MobileNetV3).

5.4 EXPERIMENT 3: SYNTHETIC 2D CONTEXT-BIAS TOY

Data: three-class 2D Gaussian blobs with four contexts A–D. Underlying class distributions are identical across contexts. The teacher is a two-layer MLP trained to high accuracy, then its logits are perturbed with context-specific scale and bias to induce margin variation (example scales $[0.5, 1.0, 1.5, 2.0]$, biases $[-0.3, 0.0, 0.2, 0.4]$). Student: two-layer MLP trained with KD (and optionally cross-entropy). KD variants: vanilla, CCAD without w_{ctx} , CCAD without T_t , full CCAD, and student-driven baselines analogous to IPWD/weighted soft labels. Metrics: overall and per-context accuracy, worst-group accuracy, ECE and NLL, per-context reliability diagrams, early stability via KD-loss variance and gradient-norm coefficient of variation, and distribution alignment via per-context ECDFs of student versus teacher margins.

5.5 FAIRNESS AND RELIABILITY

We use identical data pipelines and schedules across methods. For TTM/WTM, we grid-search global temperatures or schedules; for IPWD and weighted soft labels, we use recommended settings

with small grids; for GroupDRO, we reuse discovered contexts. We report mean $\pm$ standard deviation across multiple seeds and conduct paired t-tests on worst-group accuracy and ECE ($p < 0.05$). We measure training-time overhead with CUDA events and target under 3 percent overhead.

5.6 SCOPE AND EXTENSIONS

The same CCAD machinery applies to detection and segmentation by defining contexts at the region level and to multimodal KD via environmental embeddings. CCAD complements feature-level and information-theoretic KD [chung2020feature, ahn2019variational], comprehensive KD with causal interventions aut (b), multi-level logit transfer aut (e), dataset distillation under domain shift aut (d), and anytime adaptation aut (a).

6 RESULTS

6.1 VERIFICATION STATE

The provided execution artifact provisions the environment and integrates CCAD end-to-end: context discovery (augmentation logs plus style clustering), ECDF construction with per-context and global quantile sketches, rank-based intra-context weighting, inter-context DRO via exponentiated gradients, context-adaptive temperature, and the optional logit-side quantile equalization. Stability safeguards—warm-up ECDF collection, clipping of per-sample weights and group factors, and gradient-norm clipping—are implemented. The synthetic mechanism study instantiates explicit context-dependent teacher biases; CIFAR-100 and ImageNet-100 pipelines are assembled with metrics for accuracy, calibration, worst-group evaluation, tail-quantile accuracy, and robustness.

6.2 QUANTITATIVE OUTCOMES

The available logs do not contain numerical metrics from completed training runs on real datasets or baseline comparisons. In line with our reporting standards, we do not present accuracies, calibration errors, robustness scores, ablation tables, or statistical tests here. The absence of recorded values also precludes confidence intervals and aggregate summaries.

6.3 QUALITATIVE OBSERVATIONS

- **Feasibility and overhead.** ECDF queries and context bookkeeping add negligible complexity to the KD loop and impose no test-time overhead. Per-sample temperatures and weights integrate cleanly with batched KL computations. The optional affine equalization adds modest cost only when enabled after warm-up.
- **Early stability.** Teacher-only, rank-based controls combined with warm-up and clipping avoid the instabilities often observed with student-driven reweighting in early epochs [niu2022respecting, zhou2021ethinking]. The code instruments KD — loss variance and gradient — norm variation to quantify this once full runs are executed. **Mechanism plausibility.** The synthetic dependent margins skew for which ECDF-based equalization and adaptive temperature are redesigned; this setting directs group accuracy and calibration improvements attributable to teacher — side controls.

6.4 PLANNED COMPARATIVE EVALUATION

We will run the full CIFAR-100 and ImageNet-100 experiments with baselines: vanilla KD, TTM/WTM Zheng & Yang (2024), IPWD and weighted soft labels [niu2022respecting, zhou2021ethinking], and GroupDRO — only. Metrics will include clean accuracy, ECE, NLL, worst group accuracy over discovered contexts, tail — quantile accuracy ($q_t \leq 0.1$), and robustness metrics on ImageNet-C (mCE, ECE-C, mFR, AUROC, calibration shift). We will report ablations (remove w_{ctx} , α_t , T_t ; toggle the affine equalization) and sensitivities to β , λ , γ , and the number of contexts. Statistical significance will be assessed with paired t-tests over multiple seeds.

6.5 LIMITATIONS

Without numerical results, we cannot claim empirical superiority. Context discovery depends on feature quality; we mitigate this with hierarchical backoff and minimum-count thresholds. We will ensure fair hyperparameter tuning across all baselines and adhere to identical training pipelines to isolate the effect of CCAD’s components.

6.6 EXPECTED OUTCOMES AS HYPOTHESES

Based on design and prior analyses, we hypothesize that CCAD will: (1) improve worst-group accuracy by reallocating gradient mass toward hard tails within biased contexts, (2) enhance calibration via context-adaptive temperature, and (3) maintain or improve average accuracy, especially in cross-architecture transfer where teacher sharpness mismatches student capacity Cho & Hariharan (2019). Robustness improvements on ImageNet-C and smaller calibration shifts are anticipated. These expectations will be validated and quantified in the full runs with proper statistics.

7 CONCLUSION

We addressed context-conditional distribution mismatch in knowledge distillation and proposed Context-Conditional Adaptive Distillation (CCAD), a teacher-only, rank-based framework that equalizes per-context teacher margin distributions, allocates learning mass to worst groups, and adapts teacher sharpness to context difficulty. CCAD operates via per-context ECDFs that drive rank-stable intra-context weights, inter-context DRO weights, and a context-adaptive teacher temperature, with an optional logit-side quantile equalization. A two-stage schedule and safety mechanisms ensure early stability, rare-context smoothing, negligible training overhead, and zero test-time cost.

We provided complete implementation utilities and a rigorous evaluation plan on CIFAR-100 and ImageNet-100, including robustness suites, ablations, sensitivity analyses, and fairness safeguards. The current artifact validates feasibility and measurement instrumentation but contains no numerical results; accordingly, we refrain from efficacy claims. The next step is to execute the full suite with baselines—vanilla KD, TTM/WTM Zheng & Yang (2024), IPWD and weighted soft labels [niu2022, respecting, zhou2021, ethinking]—and GroupDRO—only, and to report statistically significant gains in worst-group accuracy and calibration with stable training and small level contexts and to multimodal KD via environmental embeddings, and combine it with feature-map and information-theoretic distillation for complementary benefits [chung2020, feature, ahn2019, variational], also level distillation, dataset distillation, and anytime adaptation [author_year, comprehensive, author_year, multi, author_year, ea].

This work was generated by AIRAS (Tanaka et al., 2025).

REFERENCES

- Anyda: Anytime domain adaptation. a.
- Comprehensive knowledge distillation with causal intervention. b.
- Knowledge distillation with auxiliary variable. c.
- Large scale dataset distillation with domain shift. d.
- Multi-level logit distillation. e.
- Sungsoo Ahn, Shell Xu Hu, Andreas Damianou, Neil D. Lawrence, and Zhenwen Dai. Variational information distillation for knowledge transfer. 2019.
- Qingyao Ai, Keping Bi, Cheng Luo, Jiafeng Guo, and W. Bruce Croft. Unbiased learning to rank with unbiased propensity estimation. 2018. doi: 10.1145/3209978.3209986.
- Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. 2019.
- Jang Hyun Cho and Bharath Hariharan. On the efficacy of knowledge distillation. *ICCV 2019*, 2019.

- Inseop Chung, SeongUk Park, Jangho Kim, and Nojun Kwak. Feature-map-level online adversarial knowledge distillation. 2020.
- Hao-Zhe Feng, Zhaoyang You, Minghao Chen, Tianye Zhang, Minfeng Zhu, Fei Wu, Chao Wu, and Wei Chen. Kd3a: Unsupervised multi-source decentralized domain adaptation via knowledge distillation. 2020.
- Bijan Mazaheri, Siddharth Jain, and Jehoshua Bruck. Robust correction of sampling bias using cumulative distribution functions. 2020.
- Yulei Niu, Long Chen, Chang Zhou, and Hanwang Zhang. Respecting transfer gap in knowledge distillation. 2022.
- Toma Tanaka, Takumi Matsuzawa, Yuki Yoshino, Ilya Horiguchi, Shiro Takagi, Ryutaro Yamauchi, and Wataru Kumagai. AIRAS, 2025. URL <https://github.com/airas-org/airas>.
- Jianfei Yang, Xiangyu Peng, Kai Wang, Zheng Zhu, Jiashi Feng, Lihua Xie, and Yang You. Divide to adapt: Mitigating confirmation bias for domain adaptation of black-box predictors. 2022.
- Kaixiang Zheng and En-Hui Yang. Knowledge distillation based on transformed teacher matching. 2024.
- Helong Zhou, Liangchen Song, Jiajie Chen, Ye Zhou, Guoli Wang, Junsong Yuan, and Qian Zhang. Rethinking soft labels for knowledge distillation: A bias–variance tradeoff perspective. 2021.